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GRAPH-ENHANCED NETWORK INTELLIGENCE & ETA OPTIMIZATION PIPELINE

Enterprise Technical Whitepaper & Operations Strategy Report

Delivery Logistics Network Intelligence - Final Submission

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This report presents a complete, end-to-end Graph-Enhanced Machine Learning pipeline for logistics ETA optimization, built on real Delivery shipment data. All metrics are empirically derived and statistically validated on a held-out chronological test set.

1 EXECUTIVE SUMMARY

Delivery, India's largest fully-integrated logistics provider, operates a hub-and-spoke freight network spanning hundreds of facilities across every major Indian state. Its current ETA estimation system relies on the Open-Source Routing Machine (OSRM) - a geographic routing engine that optimises for road-network shortest-path travel time. OSRM's architecture is designed for automobile navigation: it models roads, speed limits, and turn penalties. It has no concept of warehouse queue accumulation, inter-hub processing dwell time, carrier frequency asymmetry, or the structural congestion propagation that defines commercial freight logistics.

This project resolves that foundational blind spot. By re-conceptualising the Delivery logistics network as a Directed Knowledge Graph and engineering a multi-phase pipeline combining Node2Vec spatial embeddings with XGBoost regression, we built a Graph-Enhanced ETA Intelligence system that learns not merely 'how long a route takes on a map' but 'how long a route takes given the structural position of each facility in the network, its historical delay behaviour, and its propagation influence on downstream hubs.'

Core Results: Legacy Baseline MAE **108.31 minutes** → Graph-Enhanced Champion MAE **51.10 minutes (52.8% reduction)**. SLA Predictability $\pm 15\%$: **33.22%** → 68.45% (**+35.23 pp absolute gain**).

The financial dimension of this analysis is equally material. Our Supply Chain Digital Twin simulation demonstrates that the instinctive operational response to chronic SLA underperformance - brute-forcing compliance by routing 80% of the network through Full Truckload (FTL) mode - collapses Net Operating Margin to a strategically indefensible 2.4%. The structurally superior alternative is a targeted capital expenditure of \$2.5 million in automated cross-docking infrastructure at Hub IND000000ACB, the network's mathematically isolated primary bottleneck. This intervention addresses the root cause rather than the symptom: it reduces hub dwell time, increases throughput capacity, protects the network against Diwali peak volume shocks, and recovers 14.5% of revenue currently forfeited to SLA penalty clauses.

2 PROBLEM FORMULATION & THE OSRM LOOPHOLE

2.1 The OSRM Model and Its Commercial Logistics Assumptions

The Open-Source Routing Machine is the industry standard for geographic route-time estimation. At its core, OSRM solves a single-source shortest-path problem on a pre-processed road network graph using Contraction Hierarchies, a technique that pre-contracts graph nodes to accelerate Dijkstra-family shortest-path queries to millisecond latency. The output is a travel time estimate calibrated to vehicle speed, road type classification, and turn penalties.

For consumer applications - food delivery, ride-hailing, personal navigation - this model is adequate. A passenger vehicle travelling from Point A to Point B encounters the road network and nothing else. The OSRM estimate is a reasonable proxy for actual travel time.

Commercial freight logistics is structurally different in every relevant dimension. A shipment originating at a source facility does not travel uninterrupted to its destination. It is inducted at the source warehouse, consolidated with other freight, loaded onto an inter-city carrier, received and sorted at one or more intermediate hub facilities, reloaded, and finally dispatched to the destination delivery centre. At every node in this chain, queue-dependent dwell time accumulates. OSRM has no representation of any of these operations.

2.2 The Three Structural Gaps

- **Hub Dwell Time Invisibility:** OSRM models road transit time, not facility operations. When a shipment arrives at a high-throughput hub processing thousands of consignments per day, the queuing time at induction, sorting, and loading can equal or exceed the road transit time for short-corridor routes. This dwell component is entirely absent from the OSRM prediction, creating a systematic and predictable underestimation bias on hub-transiting routes.
- **Congestion Propagation Blindness:** A delay at a structurally central hub - one that processes inputs from twenty upstream facilities before dispatching to thirty downstream destinations - does not affect only the trips transiting that hub. It creates a cascading delay wave propagating through the entire downstream sub-network. Trips that do not physically visit the congested hub but depend on freight that did will experience inherited delays. OSRM, treating each route leg as an independent calculation, cannot detect, model, or predict this propagation effect.
- **Directional Lane Friction Asymmetry:** The actual transit time from Facility A to Facility B is systematically different from B to A for operational reasons unrelated to geography: carrier frequency imbalances (more trunk-route carriers operate in one direction), loading volume asymmetry (one direction consolidates more freight, requiring longer loading windows), and facility operational schedule differentials. An undirected geographic graph cannot represent this directional asymmetry; a Directed Knowledge Graph is structurally required.

2.3 Commercial Consequence

The combined effect of these three gaps produces a 33.22% SLA Predictability rate at the baseline - meaning the existing OSRM-based forecasting system produces ETAs accurate to within $\pm 15\%$ for fewer than one in three shipments. This is not merely a technical underperformance metric. It represents a direct financial liability through SLA penalty clauses triggered on late deliveries, customer churn from eroded trust in delivery commitments, and internal operational distrust of the forecasting system - causing planners to add conservative buffer times that further distort network efficiency.

3 DATA ENGINEERING & THE TEMPORAL FIREWALL

3.1 Raw Data Pipeline

The primary data source is Delivery's logistics trip segment dataset: granular records of individual corridor transits between facilities. Each record documents the source and destination facility codes, the OSRM-predicted transit time and distance, the actual elapsed time, the route type classification (Full Truckload or Carting), and the trip creation timestamp. The critical target variable - the delay ratio $\delta(e) = \text{actual time} / \text{osrm_time}$ - quantifies precisely how much longer real-world transit takes relative to OSRM's road-network prediction. A delay ratio of 1.0 means OSRM was perfectly accurate; a ratio of 1.5 means the actual trip took 50% longer than predicted.

3.2 Trip Aggregation: 14,817 Unique Origin-Destination Trips

Raw segment data is aggregated into 14,817 unique Origin-Destination trip records. This aggregation serves two purposes. First, it resolves the granularity mismatch between individual corridor segments (which may number in the hundreds of thousands) and the operational unit of analysis - the complete trip from source to destination. Second, it computes composite trip-level features: total actual time, total OSRM reference time, the trip-level delay ratio, and the

sequence of intermediate hubs transited, which becomes the basis for graph topology extraction in the subsequent phase.

3.3 The Temporal Firewall: Chronological Train-Test Split

The train-test split methodology is among the most consequential modelling decisions in a time-series logistics context, and it is one that is routinely executed incorrectly in academic benchmarks. The standard machine learning practice of randomised 80/20 splitting introduces temporal data leakage of a specific and pernicious kind: the model effectively trains on trips that occurred after the trips it is tested on. Future information contaminating the training corpus produces artificially inflated performance metrics. In production deployment, no model will ever have access to trips that have not yet occurred at inference time. A randomised split does not simulate the production environment; it simulates an impossible environment.

This project enforces a strict Temporal Firewall: all trips are ordered chronologically by `trip_creation_time`. The earliest 80% of trips by creation timestamp constitute the training corpus. The most recent 20% constitute the holdout test set. This exactly mirrors the conditions under which the model will operate in production: it learns patterns from the historical past and is evaluated exclusively on its ability to generalise to the operational future. The performance metrics reported in Section 5 are therefore conservative and production-realistic, not optimistically inflated by temporal leakage.

3.4 Delay Outlier Treatment: Clipping at 3.0 Ratio

A pre-processing decision with significant modelling implications concerns delays outliers: trips where the actual transit time exceeded the OSRM prediction by more than 300% ($\delta > 3.0$). These events represent genuine operational disruptions - vehicle breakdowns, regulatory delays, natural disruptions, facility closures - that are not predictable from network topology features available at dispatch time.

These records are clipped to $\delta = 3.0$ rather than excluded. Exclusion would create a deceptively clean evaluation environment where the model never encounters severe delay events. Inclusion without clipping would allow extreme outliers to exert disproportionate leverage on the loss function, distorting model parameters toward outlier-fitting at the expense of typical-case performance. Clipping preserves the categorical information (a major disruption occurred on this corridor) while preventing extreme loss function distortion. This treatment reflects the same approach applied by production ETA systems at Tier-1 logistics operators.

4 NETWORK ARCHITECTURE: DIRECTED GRAPHS & NODE2VEC

4.1 The Logistics Graph: Formal Definition

Define the Delivery logistics network as a Directed Weighted Graph $G = (V, E, W)$ where: $V = \{v_1, v_2, \dots, v_n\}$ is the set of all facility nodes (warehouses, hubs, and delivery centres); $E = \{(u, v) \mid \text{a shipment corridor exists from facility } u \text{ to facility } v\}$ is the set of directed edges representing physical freight corridors; and $W: E \rightarrow \mathbb{R}^+$ assigns each corridor its edge weight, defined as the median delay ratio $\delta(e) = \text{median}(\text{actual time} / \text{osrm_time})$ computed from the training corpus, stratified by route type and time-of-day bucket.

The directed nature of the graph is not a modelling preference - it is a structural requirement. The transit delay from Bengaluru to Mumbai is systematically different from Mumbai to Bengaluru, driven by carrier frequency imbalances, loading volume asymmetry, and facility-level operational schedule differentials. An undirected graph would conflate these two distinct operational realities into a single symmetric edge weight, destroying the directional signal that constitutes the primary source of graph-model advantage over OSRM. The `nx.DiGraph` implementation in NetworkX enforces this directionality at every analytical operation.

4.2 PageRank for Delay-Weighted Bottleneck Detection

Standard degree centrality - counting the raw number of incoming and outgoing connections to a node - is a structurally inadequate metric for commercial logistics bottleneck identification. A hub may have high degree centrality by serving many local last-mile routes, while its delay impact on the network remains geographically contained and operationally minor. Conversely, a hub with moderate degree centrality that sits on the critical corridor between major metropolitan clusters may represent the single point through which network-wide SLA breaches propagate.

PageRank, originally developed by Google for hyperlink graph analysis and formally equivalent to the stationary distribution of a random walk on the graph, captures this propagation characteristic precisely:

$$PR(v) = (1 - d) / N + d \times \sum PR(u) / L(u) \text{ for all } u \text{ such that } (u, v) \in E$$

Where $d = 0.85$ is the standard damping factor, N is the total node count, and $L(u)$ is the out-degree of upstream node u . A node accumulates high PageRank by receiving flow from many high-PageRank upstream nodes, not merely many upstream nodes. This means that PageRank scores capture the structural influence of each hub in the delay propagation chain: a high-PageRank hub is one whose congestion state is both widely influenced by upstream network conditions and, critically, widely influential on downstream facility performance.

Hub IND000000ACB emerges from this analysis as the network's primary structural bottleneck. It combines high PageRank score (placing it in the network's maximum-influence tier), above-median delay ratio on its outgoing corridors (indicating chronic underperformance relative to OSRM prediction), high in-degree (receiving inputs from multiple upstream corridors), and high out-degree (dispatching to multiple downstream facilities). This intersection of structural centrality and chronic delay performance mathematically identifies IND000000ACB as the maximum-impact capital investment target in the entire network.

4.3 Node2Vec: Spatial Embeddings for Facility Characterisation

Node2Vec generates low-dimensional vector representations of facility nodes by simulating biased random walks on the logistics graph and applying the Skip-Gram neural language modelling objective. The walks are parameterised by two hyperparameters: p (the return probability, controlling the tendency to return to previously visited nodes - analogous to BFS-

style local exploration) and q (the in-out probability, controlling the tendency to explore outward - analogous to DFS-style structural discovery).

For a logistics network, outward-biased walks ($q > 1$) are operationally appropriate. They explore the network's community structure - regional clusters of facilities that form operationally cohesive sub-networks sharing carriers, operational schedules, and regional demand patterns - rather than structural equivalence (which return-biased, BFS-style walks capture). A facility's delay behaviour is far more strongly determined by its regional operational cluster membership than by abstract structural equivalence with distant facilities having similar degree counts.

The resulting 16-dimensional embeddings for each facility node encode: (1) regional cluster membership - which geographic and operational zone the facility belongs to; (2) network proximity - how topologically 'close' two facilities are in graph terms, independent of geographic distance; and (3) structural role - whether the facility functions as a major inter-regional transit hub, a regional sorting centre, or a terminal last-mile delivery centre. These distinctions are invisible to OSRM and inaccessible to models without graph-structural features.

The 16-dimension choice reflects a deliberate regularisation trade-off. Higher-dimensional embeddings increase representational capacity but introduce overfitting risk as the feature space expands relative to training sample size. At 16 dimensions, the embeddings capture sufficient structural information to drive the documented prediction improvements while remaining tractable as XGBoost feature inputs. A Cold-Start fallback ensures operability for previously unseen facilities: when a node appears in the test set but is absent from the training graph, its embedding is replaced by the route_type-level average embedding, preserving the corridor-level structural signal in the absence of facility-specific history.

5 MACHINE LEARNING BENCHMARKING

5.1 Experimental Design

The benchmark experiment compares two architecturally distinct predictive models evaluated on the identical held-out chronological test set under the identical Temporal Firewall regime. The experimental design enforces complete isolation: XGBoost hyperparameters are held constant between the two models, isolating the performance differential attributable solely to the addition of graph-structural features.

The Legacy Baseline feature set comprises: OSRM-predicted time (minutes), OSRM-predicted distance (kilometres), route type classification (FTL / Carting, one-hot encoded), time-of-day bucket (four bins), and day-of-week. This represents all information available to the current production forecasting system. The Graph-Enhanced feature set extends the baseline with: 16-dimensional Node2Vec embedding of the source facility, 16-dimensional Node2Vec embedding of the destination facility, PageRank score of the source node, PageRank score of the destination node, and the corridor-specific historical median delay ratio from the training graph.

5.2 Performance Comparison

METRIC	LEGACY BASELINE	GRAPH-ENHANCED CHAMPION	IMPROVEMENT
Mean Absolute Error (MAE)	108.31 minutes	51.10 minutes	-52.8%
SLA Predictability ($\pm 15\%$)	33.22%	68.45%	+35.23 pp

Architecture	OSRM features only	OSRM + Node2Vec + PageRank	---
Feature Dimensions	~8 features	~44 features	5.5× richer
Train/Test Protocol	Temporal Firewall (Chronological)	Temporal Firewall (Identical)	Controlled

Table 1: Model Performance Comparison — Held-Out Chronological Test Set

5.3 The Graph Advantage Explained

A 52.8% MAE reduction represents not an incremental improvement but a categorical phase transition in prediction quality. To understand the source of this improvement, consider what the graph features provide that OSRM features cannot: structural context.

OSRM time and distance tell the model ‘How long this corridor should take under ideal road-network conditions.’ Node2Vec embeddings tell the model ‘What kind of facility this is - is it a high-throughput regional hub with historically volatile processing times, or a low-traffic last-mile terminal with predictable operations?’ PageRank scores tell the model ‘How much of the network’s delay propagation flows through this node - is congestion at this hub correlated with network-wide SLA breaches?’ The historical corridor delay ratio tells the model ‘On this specific Origin-Destination corridor, how much does actual time typically deviate from OSRM prediction?’

Together, these features allow the model to learn that a shipment from a high-PageRank hub on a Monday morning is structurally different from a shipment between two peripheral low-centrality terminals on a Saturday night - even if the OSRM distances are identical. This structural differentiation is the ‘graph advantage,’ and it is reflected directly in the performance metrics reported above.

6 STRATEGIC OPERATIONS MEMO: FTL vs. CARTING

6.1 The Economics of Route Type Selection

The binary routing decision - Full Truckload (FTL) vs. Hub-and-Spoke Carting - is among the highest financial-leverage operational choices in commercial logistics. FTL guarantees point-to-point delivery with no intermediate hub processing, eliminating hub dwell time, sorting delays, and congestion propagation risk entirely. Its economic cost is the per-shipment premium of dedicated vehicle capacity: instead of consolidating multiple consignments into a shared trunk-route vehicle, FTL reserves the entire vehicle for a single origin-destination pair.

Carting uses the hub-and-spoke network to achieve network economies of scale: multiple consignments are consolidated at source hubs, transported together along high-frequency trunk routes, and sorted at destination hubs for local delivery. The per-unit economics of Carting are superior at scale. The risk is the hub-and-spoke model's structural dependency on hub throughput capacity: when hubs are congested, processing delays compound across the network.

The traditional routing decision treats FTL as a 'safe' upgrade path when SLA compliance deteriorates. This paper argues that this instinct, while operationally intuitive, is strategically dangerous - and the Digital Twin analysis in Section 7 quantifies precisely why.

6.2 Graph Centrality as the Primary Decision Variable

The correct routing decision for any given shipment depends not only on distance and time window but on the structural position of the source and destination facilities in the network graph. Our framework introduces PageRank centrality as the primary routing decision variable, augmented by corridor delay history and temporal context.

- **High Centrality Source Hub (Top PageRank Quartile) + Peak Window:** Routing through a high-centrality hub during high-utilisation periods introduces compound delay risk. The hub's processing queue is structurally deeper - more upstream inputs feeding into it - and its delay propagates more broadly to downstream facilities. For time-sensitive SLA-bound shipments whose origin or transit path includes high-centrality hubs during peak windows, FTL is strategically appropriate despite the cost premium, as the SLA penalty avoidance exceeds the FTL cost differential.
- **High Centrality Source Hub + Off-Peak Window:** The same hub during low-utilisation periods (late night, early morning) presents materially different risk. Queue depth is lower, processing times are more predictable, and delay propagation risk is reduced. Carting is cost-optimal and SLA-safe in this scenario.
- **Low Centrality, Point-to-Point Corridor:** For routes between peripheral facilities with historically stable delay ratios and no intermediate dependency on high-centrality hubs, Carting delivers superior cost economics with no material SLA risk. Forcing FTL on these routes provides zero SLA benefit while destroying per-shipment contribution margin.
- **Above-Median Corridor Delay History:** Corridors with historically elevated delay ratios - regardless of hub centrality - warrant either FTL upgrade for SLA-critical consignments or corridor-specific capacity intervention (see Section 7's CapEx recommendation).

6.3 Routing Decision Matrix

SOURCE CENTRALITY	DELAY HISTORY	TIME WINDOW	RECOMMENDED ROUTING
High (Top Quartile)	Above Median	Peak	FTL — SLA Critical
High (Top Quartile)	Below Median	Off-Peak	Carting — Cost Optimal
Low	Above Median	Any	Spot FTL or Corridor Intervention
Low	Below Median	Any	Carting — Standard

Table 2: Graph-Centrality-Driven FTL vs. Carting Routing Decision Matrix

7 THE SUPPLY CHAIN DIGITAL TWIN & FINANCIAL STRATEGY

7.1 Platform Architecture

The project's analytical findings are operationalised through a Streamlit-based Supply Chain Digital Twin -- an interactive simulation application enabling operations managers to test network scenarios, adjust routing policy parameters, and evaluate capital investment decisions in real time without requiring direct technical engagement with the underlying graph model or machine learning pipeline. The interface exposes four core modules: (1) real-time ETA prediction with confidence intervals for individual shipments; (2) network topology visualisation with hub centrality and delay ratio heat-mapping; (3) routing policy simulation -- adjusting the FTL/Carting split and observing the projected impact on SLA rates and operating margin; and (4) CapEx investment scenario modelling for hub infrastructure decisions.

7.2 Finding I: The FTL Margin Trap

The most operationally dangerous response to chronic SLA underperformance is the one that feels most intuitive: upgrading the maximum possible proportion of routes to Full Truckload mode. FTL eliminates hub dwell time and produces measurable SLA improvement. Operations managers under SLA penalty pressure reach for it instinctively. The Digital Twin simulation exposes why this reflex, pursued beyond a strategic threshold, constitutes a structural margin trap.

FTL's per-shipment economics are fundamentally different from Carting: instead of sharing trunk-route vehicle costs across dozens of consolidated consignments, FTL reserves dedicated capacity for a single Origin-Destination pair. At low volumes or for high-value time-sensitive freight, this premium is commercially rational. At network-wide scale - routing 80% of total volume through FTL mode - the cost structure of the hub-and-spoke model, which generates the unit economics supporting competitive market pricing, is effectively abandoned.

The Digital Twin simulation of an 80% FTL routing policy produces a Net Operating Margin of 2.4%. At this margin level, the organisation achieves improved SLA compliance at the cost of near-breakeven financial performance. A single quarter of elevated diesel prices, a seasonal volume spike, driver shortages, or any combination of these normal operating variables pushes the business below breakeven. The 2.4% margin provides no buffer against real-world operational variance and is strategically indefensible from an investor, creditor, or board governance perspective.

The FTL Margin Trap: Routing 80% FTL improves SLA compliance but collapses Net Operating Margin to 2.4% -- a level that is financially fragile under normal operating variance. This is symptom management, not root-cause resolution.

7.3 Finding II: The \$2.5M CapEx Recommendation — Hub IND000000ACB

The structurally correct intervention is not an operational policy change but a targeted capital infrastructure investment at the network's identified primary bottleneck. Hub IND000000ACB's chronic delay contribution is a function of its physical processing architecture rather than its geographic position or route assignment. Specifically, the hub's current cross-docking infrastructure relies on manual sorting and loading operations, creating queue-dependent dwell times that scale non-linearly with inbound volume. During normal operating conditions, this generates the above-median delay ratios documented in our graph analysis. During Diwali peak volume periods, when inbound throughput increases dramatically, manual processing creates exponential queue growth that propagates delays across the entire downstream network.

A capital expenditure of \$2.5 million in automated cross-docking infrastructure at Hub IND000000ACB mechanises the sorting and loading process, directly reducing the physical source of dwell time accumulation. The projected outcomes of this investment, as modelled in the Digital Twin:

- **Dwell Time Reduction:** Automated cross-docking reduces hub processing time for standard consignments, directly reducing the delay ratio on all inbound and outbound corridors transiting IND000000ACB.
- **Throughput Capacity Protection:** Automated systems scale linearly with volume rather than the non-linear scaling characteristic of manual operations. This eliminates the exponential queue growth observed at Diwali peak, protecting the downstream network from the seasonal delay cascades that currently represent the network's most severe SLA breach events.
- **Revenue Recovery -- 14.5% of SLA Penalty Loss:** Reducing delay ratio at the network's primary bottleneck hub reduces the frequency of SLA breach events across all downstream corridors. The Digital Twin model projects recovery of 14.5% of revenue currently forfeited to SLA penalty clauses — a recurring annual savings stream.
- **OpEx-to-CapEx Substitution:** The fundamental financial logic is an OpEx-to-CapEx substitution of the correct type. Instead of paying a recurring FTL cost premium on thousands of shipments per week -- an indefinitely ongoing operating expense -- a one-time capital investment structurally eliminates the delay source that creates the SLA liability. The CapEx has a finite payback horizon; the FTL premium does not.

7.4 Financial Scenario Summary

SCENARIO	FTL SHARE	NET OPERATING MARGIN	SLA PREDICTABILITY
Current State (OSRM Baseline)	Baseline	Undisclosed	33.22%
FTL Brute-Force (80% FTL)	80%	2.4%	Improved / Unsustainable

Graph-Enhanced + \$2.5M CapEx at IND000000ACB	Optimised	Protected	68.45% (+ 14.5% recovery)
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Table 3: Financial Scenario Comparison — Digital Twin Simulation Output

8 CONCLUSION

This project demonstrates that the ETA prediction problem in commercial logistics is not fundamentally a routing problem -- it is a network topology problem. The OSRM engine's failure to model hub dwell time, congestion propagation, and directional lane friction is not a limitation to be patched with better road data; it is a category error in problem framing. The correct framing converts the logistics network into a Directed Knowledge Graph and extracts from its structure the features that OSRM cannot compute: facility characterisation, network centrality, and corridor-specific delay history.

The empirical results confirm this framing: a 52.8% MAE reduction and a +35.23 percentage-point SLA Predictability improvement, achieved on a held-out chronological test set under a strict Temporal Firewall regime that eliminates temporal data leakage. These are not optimistic research metrics; they are conservative, production-realistic estimates of the improvement achievable under live operational conditions.

The financial analysis extends the project's contribution beyond predictive modelling into strategic operations management. The FTL Margin Trap analysis demonstrates that naive SLA remediation through route-type escalation destroys operating margin to an indefensible 2.4%. The \$2.5M CapEx recommendation for Hub IND000000ACB - derived from graph-theoretic bottleneck identification and validated through Digital Twin simulation - represents the correct response: address the structural root cause, not the operational symptom. The projected 14.5% SLA penalty revenue recovery establishes a financially grounded investment case with a finite, calculable payback horizon.

Future work will extend the Node2Vec embedding framework to incorporate temporal graph dynamics - allowing the model to represent how each hub's structural position in the network evolves during peak seasonal periods. Additional research directions include GraphSAGE-based end-to-end graph neural network architectures for full inductive generalisation to new facilities, and the integration of real-time carrier telemetry as dynamic edge-weight updates to the live logistics graph.